Towards Robust Diabetes-Specialized LLMs: Reflection- and Curriculum-based Instruction Tuning Across Diverse Tasks

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Abstract

Objective: Diabetes is a globally prevalent and complex chronic condition that requires continuous interpretation of glycemic trends, personalized lifestyle interventions, and sustained patient education. While large language models (LLMs) have shown promise in various biomedical applications, existing general-purpose and biomedical LLMs often underperform in diabetes-specific reasoning and instruction-following tasks. To address this limitation, we present a diabetes-specialized LLM fine-tuned on a curated instruction dataset that reflects the nuances of diabetes management.

Methods: Our approach integrates two reflection-based replacement metrics—Instruction-Following Difficulty (IFD) and reversed-IFD (r-IFD)—to evaluate the model's receptiveness to instructions and to filter and substitute suboptimal prompts. In addition, we apply a curriculum instruction tuning strategy that sequences instructions from easier to more challenging based on model sensitivity, mitigating catastrophic forgetting while promoting stable acquisition of domain expertise.

Results: We evaluate the model across multiple diabetes-related task categories, including question answering, natural language inference, information extraction, summarization, generation and diabetes-friendly dietary recommendation. Our model demonstrates superior performance in accuracy compared to strong biomedical baselines, and generates meal plans that are 0.66% more suitable than those produced by GPT-4, as measured by the The Diet Quality Index-International(DQI-I) score—an international metric for nutritional quality.

Conclusion: These results highlight the potential of domain-specific instruction tuning to transform general-purpose LLMs into expert-level assistants for diabetes management.

Key words: Diabetes Management, Reflection-based Replacement, Instruction Following Difficulty, Reversed-Instruction Following Difficulty, Curriculum Instruction Tuning, diabetes-friendly Nutrition

Introduction

Diabetes is one of the most prevalent and complex chronic diseases globally, currently affecting more than 500 million individuals and contributing significantly to the global burden of non-communicable diseases. According to a comprehensive pooled analysis of 1108 population-representative studies including over 141 million participants, the worldwide prevalence of diabetes has risen continuously from 1990 to 2022, and it is projected to escalate even further in the next decades[25]. Managing diabetes effectively requires not only pharmacological treatment but also ongoing interpretation of blood glucose fluctuations, dietary adaptations, personalized lifestyle interventions, and continuous patient education. As the demand for scalable, patient-centered care rises, the healthcare community increasingly seeks robust AI tools, such as large language models (LLMs), that can support clinical decision-making and self-management guidance.

LLMs have recently gained traction in the biomedical domain due to their capability to generate human-like responses and synthesize complex information. In the context of diabetes management, large language models (LLMs) have been leveraged across a diverse set of tasks tailored to patient needs. For instance, integrated systems combining image-based data with LLM-generated responses have been shown to enhance interpretability in primary management settings[11]. Retrieval-augmented generation (RAG) architectures have been employed to support diabetes education by grounding LLM outputs in trusted reference sources, thereby improving factual reliability[17]. In another example, large language models have been applied to analyze CGM data directly and summarize trends over time for patients and clinicians[1]. But, there is no practical diabetes management LLM that answers questions about diabetes and even provides diet recommendations.

Also, these promising directions still face critical limitations. Biomedical LLMs such as BioBERT[5] and BioGPT[12] have demonstrated high performance on general biomedical QA benchmarks but often underperform on reasoning tasks related to diabetes management, which require clinical nuance, contextual understanding, and personalized judgment. This limitation was further substantiated in a recent prospective study, where even advanced biomedical LLMs failed to achieve satisfactory accuracy and alignment when evaluated on Multiple diabetes-related task. Similarly, general-purpose LLMs like ChatGPT and Bard continue to struggle with endocrinology-related queries, frequently providing vague or partially incorrect answers that lack alignment with clinical guidelines[4]. These limitations underscore the need for models that are not only trained on diabetes-specific data but also optimized for diabetes-specific reasoning and instruction-following in diabetes management, including accurate answers to diabetes-related questions and diabetes-friendly dietary recommendations.

To address this gap, we propose a diabetes-specialized large language model trained on a diabetes-specific instruction dataset designed to reflect the nuances of diabetes management. The dataset includes a diverse set of prompts covering glycemic trend interpretation, insulin dose adjustment, nutritional guidance, symptom analysis, and patient counseling. Importantly, the model is trained not merely to recall factual information but to engage in higher-order reasoning that emulates the decision-making patterns of healthcare professionals. To support this goal, we go beyond merely collecting high-quality instructions. Instead, we systematically refine the dataset to be pedagogically aligned with the model's capacity and learning dynamics. To develop effective domain-specific LLMs, it is critical not only to curate high-quality instructions but also to align them with the model's level of receptiveness. Inspired by the notion of student-guided data selection introduced in Selective Reflection-Tuning [9], we adopt a reflection-based replacement strategy that filters and substitutes instruction samples based on their pedagogical value and the model's learning capacity. In addition, to mitigate catastrophic forgetting, improve alignment with diabetes-specific instruction dataset, and enhance training robustness, we organize instruction dataset following a curriculum instruction tuning framework, where samples are sequenced from easier to more challenging tasks based on model sensitivity—a strategy motivated by the principles of data progression outlined in Data-CUBE [21]. This organization is not defined by surface-level complexity, but by how effectively the model internalizes and responds to instructional content.

To construct an diabetes-specific instruction dataset optimized for the model's learning trajectory, We employ a reflection-based replacement strategy that balances pedagogical difficulty and the model's learning capacity. Specifically, We adopt two replacement metrics—Instruction-Following Difficulty (IFD) and reversed-IFD (r-IFD)—to filter and substitute instruction samples that are either too easy or elicit vague responses. The IFD score measures how difficult a prompt is for the model to follow, while the r-IFD score captures the ambiguity of the model's response. By filtering out instructions samples that are noisy, redundant, or cognitively mismatched, and substituting them with more effective examples, we ensure the dataset is both challenging and instructionally coherent. This approach is inspired by recent advances in self-guided instruction selection and reflection-tuning frameworks [9,10] . As generating synthetic instruction datasets with models like GPT-4 becomes more common, our method addresses the overlooked issue that not all instructions are equally beneficial for fine-tuning.

To prevent catastrophic forgetting and promote stable acquisition of domain-specific knowledge, we adopt a curriculum instruction tuning strategy inspired by frameworks such as DATA-CUBE [21]. Our approach begins training with prompts that exhibit low Instruction-Following Difficulty (IFD) scores—representing tasks the model already handles competently—to reinforce foundational diabetes knowledge without overwhelming the model. As training progresses, we gradually introduce prompts with higher IFD scores, which involve more complex reasoning and domain-specific depth. This gradual escalation in instructional complexity ensures that the model builds upon prior knowledge in a structured manner while preserving previously acquired information and enhancing its performance on diabetes-specific tasks.

We evaluate our model across multiple diabetes-related task categories, including question answering (QA), natural language inference (NLI), information extraction (IE), summarization, generation, and diabetes-friendly dietary recommendation. Our model significantly outperforms baseline biomedical LLMs in accuracy. In diabetes-friendly dietary recommendation tasks, it generates meal suggestions that consider both nutrient balance and glycemic index. Notably, the diet plans produced by our model are 0.66% more suitable than those generated by GPT-4, as measured by the Diet Quality Index-International (DQI-I) score—an internationally standardized metric for assessing the overall nutritional quality and balance of a diet.

In summary, this study introduces a diabetes-specialized LLM built on LLaMA-3 and optimized using reflection-based replacement and curriculum instruction tuning strategies. Our contributions include: (1) the integration of IFD and r-IFD metrics to evaluate the model's receptiveness to instructions and to guide the filtering and substitution of suboptimal prompts (2) the implementation of a task complexity–aware curriculum instruction tuning strategy to improve model robustness and mitigate catastrophic forgetting; and (3) empirical evidence of superior performance in diabetes-related questions and diabetes-friendly dietary recommendations, consistently outperforming strong biomedical LLM baselines. These results collectively demonstrate that domain specific instruction tuning, when designed with care and optimized strategically, can successfully transform general purpose LLMs into expert level assistants for diabetes management, narrowing the gap between general capability and diabetes specialized reasoning.

Related Work

Biomedical LLMs, like BioBERT [5] and PubMedBERT [22], improve NLP tasks through domain-specific pretraining. More recent models such as BioGPT [12] and MedLLaMA [20] offer generative capabilities for medical QA. While benefiting from medical literature, these models are often general-purpose, lacking instruction tuning or disease-specific adaptation. Consequently, they underperform in specialized management domains requiring nuanced reasoning, often failing to produce accurate or safe responses in patient-facing contexts due to limited instruction tuning and alignment for complex medical queries [14].

Researchers are developing disease-specific LLMs to address these limitations. Examples include models for dementia care [14], liver disease with RAG [3], and rare disease concept normalization [16]. These models show specialized pretraining or fine-tuning can bridge the gap between general biomedical models and specific disease applications. Xie, Q. et al. [19] introduced a foundational model for comprehensive medical text analysis. However, no instruction-tuned model has yet been proposed for diabetes management, a domain needing knowledge retrieval, patient guidance, and dietary recommendations, highlighting a key research opportunity.

Instruction-tuned LLM performance depends heavily on instruction quality. Self-Instruct [15] showed scaling instruction generation improves performance, while LIMA [2] emphasized quality over quantity. Studies like 'Stronger Models are Not Stronger Teachers' [24] revealed that teacher model outputs can lead to poor alignment, especially for student models with lower capacity. This highlights the need for instruction datasets tailored to the target model's preferences, leading to approaches like Tuna [8] and LESS [18] that identify effective samples. Selective Reflection-Tuning [9] and From Quantity to Quality [10] adopted student-driven filtering. Our work extends this by introducing Instruction-Following Difficulty (IFD) and reversed-IFD (r-IFD) metrics to optimize dataset construction for learning efficiency and domain alignment.

Curriculum instruction tuning, derived from cognitive science, orders training examples from easy to challenging to improve convergence. In LLMs, TAPIR [23] and Instruction Tuning with Human Curriculum [7] apply this principle by organizing prompts based on instructional demands. Curri-DPO [13] showed strong gains by ordering preference pairs from easy to hard, improving alignment and performance. The DATA-CUBE framework [21] uses a dual-layer curriculum based on task difficulty. Our approach adapts these ideas by ranking prompts using IFD scores, constructing a curriculum from low-IFD to high-IFD tasks. This model-centered strategy ensures effective acquisition of diabetes-specific reasoning skills without compromising general biomedical competencies. We argue that optimizing instruction datasets through filtering (IFD/r-IFD) and ordering (curriculum strategies) from the student model's perspective is crucial for stronger domain-specific performance.

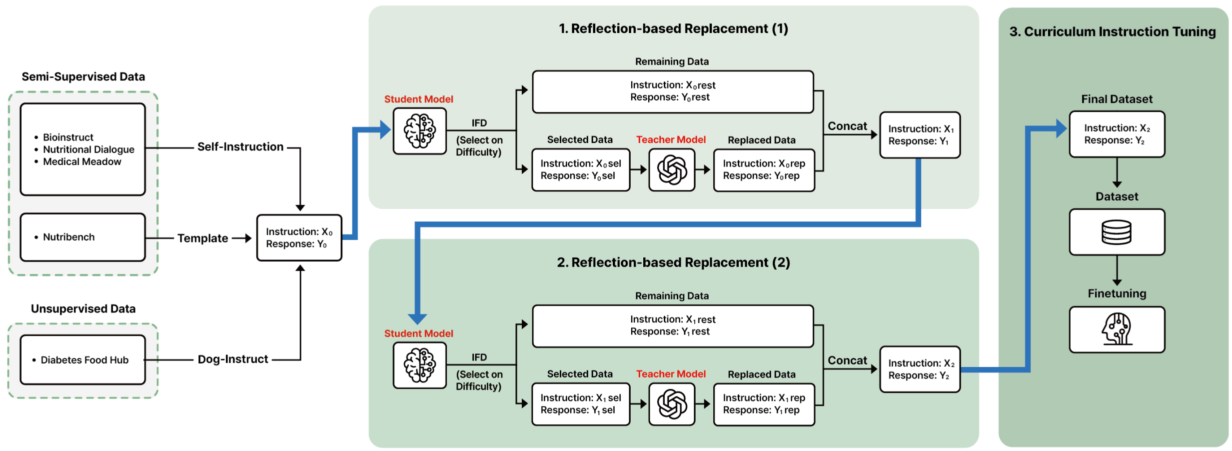

Figure 1. Architecture of diabetes-specialized LLM tuning with reflection-based replacement and curriculum instruction tuning.

Methodology

Overview

To address the limitations of general biomedical LLMs in performing diabetes-specific reasoning, instruction-following tasks specific to diabetes management, we adopt a model-centric instruction refinement approach guided by the student model's receptiveness. Instruction–response pairs are evaluated using Instruction-Following Difficulty (IFD) and reversed-IFD (r-IFD) to measure both cognitive challenge and semantic feasibility. Based on these metrics, low-quality samples are selectively substituted via teacher-assisted revision. We further apply curriculum instruction tuning by sequencing training data from easy to challenging according to IFD scores. This process enables the training of a diabetes-specialized LLM capable of accurately performing multiple diabetes-related tasks and providing diabetes-friendly dietary recommendations.

Task Definition

The primary objective of our approach is to fine-tune a large language model (LLM) to perform multiple diabetes-related tasks and provide diabetes-friendly dietary recommendations. The instruction dataset spans multiple levels of complexity, covering factual queries about diabetes mechanisms, and nutrition scenarios tailored to patient-specific profiles. Rather than targeting general biomedical reasoning, our model focuses on multiple diabetes-related tasks and diabetes-friendly dietary recommendations, aiming to serve as a specialized AI assistant for diabetes management.

Expected Input/Output

The model accepts as input a natural language instruction related to diabetes. Examples include clinical queries such as "Explain the effect of complex carbohydrates on postprandial glucose levels" and patient-specific prompts like "Suggest a breakfast plan for a patient with type 2 diabetes receiving basal insulin."

The expected output is a medically accurate and contextually appropriate response tailored to the task type. Depending on the task, the model may generate fact-based answers for question answering (QA), extract structured information for information extraction (IE), perform inference for natural language inference (NLI), produce coherent texts for generation and summarization tasks, or recommend diabetes-friendly dietary plans aligned with nutritional and glycemic considerations.

Pipeline Overview

To support this objective, we implement a two-stage instruction optimization process tailored to the diabetes domain:

Reflection-based replacement: We use Instruction-Following Difficulty (IFD) and reversed-IFD (r-IFD) scores to identify samples that are both too easy and semantically ambiguous. Such low-value samples are revised through reflection-based replacement, while the remaining examples—those with sufficient challenge and alignment—are preserved for training.

Curriculum instruction tuning: We adopt a progressive curriculum instruction tuning strategy to expose the model to instructional data in order of increasing difficulty. This helps stabilize training dynamics and ensures that fundamental domain knowledge is reinforced prior to the introduction of complex cases.

This process builds upon and extends prior work on instruction-tuning and curriculum instruction tuning in NLP. Our key contribution lies in combining reflection-based replacing—driven by IFD and r-IFD—with domain-specific curriculum instruction tuning, enabling adaptive instruction optimization for diabetes management.

Instruction Dataset Seed Construction

A critical foundation for effective instruction tuning lies in constructing a dataset optimized for model receptiveness, capturing both the structured logic of multiple diabetes-related tasks and the contextual complexity of diabetes-friendly dietary recommendations. To this end, we curated a domain-specialized instruction dataset by aggregating and transforming content from multiple public sources, using a combination of Self-Instruct, FLAN-style prompting, and Dog-Instruct techniques for semi-supervised instruction generation and revision.

We sourced our data from five key corpora: BioInstruct and Medical Meadow contributed diverse biomedical instructions covering condition definitions, symptom interpretation, and treatment recommendations; Nutritional Dialogue provided free-form conversational exchanges centered on diet-diabetes interactions; and both Diabetes Food Hub and Nutribench offered structured nutritional facts and recipe-level guidance for glycemic control. These datasets represent a rich combination of clinical expertise and practical dietary insight.

To convert this heterogeneous content into a unified instruction–response format suitable for LLM tuning, we applied a multi-stage transformation pipeline. For datasets such as BioInstruct, Nutritional Dialogue, and Medical Meadow, we adopted instruction–response pairs generated via the Self-Instruct framework or retained existing ones when already aligned. For Nutribench, we applied customized prompt templates inspired by the FLAN methodology to standardize the format for nutritional tasks. Lastly, for Diabetes Food Hub, we employed Dog-Instruct technique, which leverages back-translation and structural rewriting to transform unstructured textual content—particularly long-form or informal passages—into high-quality instruction–response examples.

The resulting instruction dataset encompasses a diverse range of task types. These include: (1) disease definition and explanation tasks (e.g., "What is diabetic ketoacidosis?"); (2) diabetes-friendly recommendation tasks tailored to diabetic conditions and symptoms (e.g., "Suggest a low-sodium meal plan for a patient experiencing diabetic nephropathy" or "Create a breakfast plan suitable for managing morning hyperglycemia"); (3) glycemic trend analysis, which requires causal reasoning over dietary inputs and physiological responses (e.g., "Why does blood sugar spike after eating rice?"); (4) diet optimization tasks that compare foods in terms of nutritional composition (e.g., "Compare the protein and fiber content of tofu and quinoa").

To ensure topical focus and domain specificity, we applied a keyword-based filtering strategy. Only samples explicitly mentioning "diabetes" were selected for inclusion in the dataset. This ensured that the instruction dataset remained tightly aligned with the target domain of diabetes management.

Collectively, this transformed dataset serves as a robust foundation for instruction-level fine-tuning, supporting both the reinforcement of diabetes-specific knowledge and the generation of diabetes-friendly dietary recommendations in downstream modeling stages.

Instruction refinement using reflection-based Replacement

Instruction datasets often vary in their pedagogical utility when viewed from the perspective of the student model: some samples provide minimal learning signal, while others present cognitively meaningful challenges that promote deeper representation learning. Motivated by this observation, we apply a reflection-based replacement mechanism to selectively filter and substitute low-value samples while preserving those with higher instructional relevance. Our approach draws inspiration from Selective Reflection-Tuning: Student-Selected Data Recycling for LLM Instruction-Tuning [9], which emphasizes the importance of student model–centric evaluation over teacher-generated data quality heuristics.

We apply a two-stage replacement pipeline guided by two complementary metrics: Instruction-Following Difficulty (IFD) and Reversed Instruction-Following Difficulty (r-IFD). From the perspective of the student model, IFD is used to filter instruction–response pairs in which the instruction is too trivial—offering little learning value because the model likely already knows the answer. In contrast, r-IFD is used to filter instruction-response pairs with vague or ambiguous responses that lack strong semantic consistency with the given instruction, potentially leading to confusion during training. Instruction–response pairs that are either too easy or too semantically unclear are replaced with revised examples to enhance pedagogical utility and consistency.

Instruction-Following Difficulty (IFD) measures how much an instruction contributes to the learning process from the perspective of the student model. Rather than quantifying difficulty in an abstract sense, IFD captures the extent to which the presence of an instruction helps the model generate a more accurate and meaningful response compared to when no instruction is given. If the model can generate the correct response even without the instruction, the instruction is likely too simple—requiring little reasoning effort and offering minimal learning value. In contrast, if the instruction significantly improves the model's ability to generate the correct answer, it suggests that the instruction provides meaningful guidance and stimulates deeper understanding. High-IFD instruction–response pairs encourage the model to go beyond rote recall and engage in more deliberate reasoning. These samples play an important role in improving instruction-following behavior and enhancing the overall quality of the model's responses by fostering cognitive effort and model-level comprehension.

$$\mathrm{IFD}_\theta(y \mid x) = \frac{\mathrm{ppl}(y \mid x)}{\mathrm{ppl}(y)} = \exp\left(\mathcal{L}_\theta(y \mid x) - \mathcal{L}_\theta(y)\right) \quad (1)$$

$$(x_1, y_1) = \arg\max_{(x,y)} \mathrm{IFD}_\theta(y \mid x) \quad (2)$$

Reversed Instruction-Following Difficulty (r-IFD) measures how clearly a generated response reflects the original instruction, from the perspective of the student model. Rather than evaluating response quality in isolation, r-IFD captures how well the instruction can be semantically inferred from the response alone. If the model can easily recover the intent or content of the instruction just by reading the response, The sample is likely to exhibit strong semantic consistency and clear instructional grounding. In contrast, if the response is vague or underspecified, making it difficult to infer the original instruction, the sample may lack clarity and fail to provide reliable learning signals. Low-r-IFD instruction–response pairs promote alignment and interpretability, allowing the model to better understand the relationship between instructions and responses. These samples help reduce ambiguity, strengthen semantic consistency, and ultimately improve the model's ability to generalize from instruction-following tasks.

$$\text{r-IFD}_\theta(x \mid y) = \frac{\text{ppl}(x \mid y')}{\text{ppl}(x)} = \exp\left(\mathcal{L}_\theta(x \mid y') - \mathcal{L}_\theta(x)\right) \quad (3)$$

$$(x_2, y_2) = \underset{(x_1,y_1)}{\arg\min}\, \text{r-IFD}_\theta(x_1 \mid y_1) \quad (4)$$

Based on the two reflection-based metrics, we retain instruction–response pairs that demonstrate both high IFD—indicating meaningful instructional contribution—and low r-IFD—reflecting strong semantic consistency. Samples that fall outside this range, such as overly simple instructions or ambiguous responses, are identified as suboptimal for learning. These are refined through a reflection-based replacement process using GPT-3.5 and structured prompts, with the goal of improving clarity, strengthening instruction–response consistency, and enhancing overall dataset quality from the perspective of the student model.

$$(x', y') = \begin{cases} \text{Replace}(x, y), & \text{if } \text{IFD}_\theta(y \mid x) \leq 0.45 \\ & \quad \text{and } \text{r-IFD}_\theta(x \mid y) \geq 0.98 \\ (x, y), & \text{otherwise} \end{cases} \quad (5)$$

By selectively recycling only those examples that provide substantial learning signals to the student model, this approach ensures that fine-tuning is guided by both task complexity and semantic consistency. This process leads to improved model performance, as it enhances instruction-following capability, promotes deeper reasoning, and ultimately improves the quality of generated responses.

Curriculum Instruction Tuning for Learning Optimization

Since our instruction dataset covers multiple task types—including question answering, information extraction, natural language inference, summarization, generation and diabetes-friendly dietary recommendation—the resulting instruction–response pairs vary widely in both structure and response length. Fine-tuning on such heterogeneous data without any ordering can destabilize training and hinder the model's ability to internalize complex tasks. To address this, we apply a staged curriculum instruction tuning strategy that sequences instructions from easier to more challenging, helping the model gradually adapt to increasing task complexity and response diversity.

We first rank all instruction–response pairs based on their Instruction-Following Difficulty (IFD) scores, which reflect the relative difficulty of each sample from the perspective of the student model. The dataset is subsequently partitioned into mini-batches arranged in increasing order of difficulty. Following the Data-CUBE approach [6], we construct each batch to minimize intra-batch variance in difficulty and train the model in a sequential order from easy to challenging.

[t] [1] Instruction dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, model θ, number of stages B Fine-tuned model via staged curriculum instruction tuning Initialize empty list $\mathcal{D}'$ each $(x, y) \in \mathcal{D}$ Compute $\text{IFD}_\theta(y \mid x)$ Append $(x, y, \text{IFD}_\theta(y \mid x))$ to $\mathcal{D}'$ Sort $\mathcal{D}'$ in ascending order of IFD scores Partition $\mathcal{D}'$ into B mini-batches: $\{\mathcal{B}_1, \mathcal{B}_2, \ldots, \mathcal{B}_B\}$ $b = 1$ to B Fine-tune model θ on batch $\mathcal{B}_b$

This progressive training approach stabilizes learning by mitigating task-level heterogeneity and reducing difficulty divergence across batches. It facilitates robust convergence and prevents catastrophic forgetting by reinforcing previously acquired general medical competencies before introducing more specialized, domain-specific instructions. As a result, the model can effectively acquire expert-level reasoning in diabetes-related tasks without compromising foundational biomedical knowledge.

Implementation Details

We fine-tuned our model using the LLaMA3 architecture, experimenting with both 3B and 8B parameter variants, made available through Hugging Face's Transformers library. To enable parameter-efficient tuning, we employed Low-Rank Adaptation (LoRA) via the PEFT framework, setting the LoRA rank to 16, scaling factor $\alpha = 16$, and dropout rate to 0.05.

The training configuration was as follows: an effective batch size of 64 was achieved through a per-device batch size of 16 combined with a gradient accumulation step of 4. We used the AdamW optimizer with a learning rate of 2×10^{-5}, and conducted training over two epochs. The maximum sequence length was set to 2048 tokens.

All experiments were performed on a single NVIDIA A100 GPU with 40GB of memory. To maintain alignment between data quality and model input, instruction filtering was performed prior to tokenization, ensuring consistent batching and curriculum alignment during training.

The complete pipeline—including reflection-based replacement, IFD-guided curriculum instruction tuning, and domain-adaptive fine-tuning—produces a lightweight yet highly specialized model. It exhibits strong instruction-following behavior across diverse diabetes-related tasks, from clinical reasoning to diabetes-friendly dietary recommendations, while effectively retaining general biomedical capabilities.

Experimental Setup

We evaluate our diabetes-specialized LLM using a set of multiple diabetes-related task categories to assess its effectiveness across multiple task types, including question answering (QA), natural language inference (NLI), information extraction (IE), summarization, generation, and diabetes-friendly dietary recommendation. To contextualize our model's performance, we conduct comparative evaluations against several state-of-the-art instruction-tuned biomedical LLMs, including MedAlpaca 7B, PMC-LLaMA, ChatDoctor, Alpacare LLaMA2 7B, BioMistral, and ME-LLaMA.

Our evaluation covers five task types: (1) question answering (QA) on MedQA-USMLE, MedMCQA, and PubMedQA; (2) natural language inference (NLI) on MedNLI; (3) information extraction (IE) on ADE Corpus; (4) summarization of medical dialogues using the NoteChat dataset; and (5) answer generation using ICliniq, a real-world doctor–patient Q&A corpus.

For classification tasks such as QA, NLI, and IE, we report accuracy (ACC). For generative tasks including summarization and generation, we compute BLEURT and BERTScore to evaluate semantic similarity. In addition, we score each generated output on a 0–1 scale to assess coherence, completeness, and naturalness using GPT-4 as an evaluator.

To further test domain specialization, we evaluate the model's ability to generate nutritionally appropriate diet plans for diabetes management. In this task, we compare our model with GPT-4, a strong general-purpose baseline. Evaluation metrics include the Diet Quality Index–International (DQI-I) score and GPT-4-based scoring of text coherence, completeness, and naturalness. For each model, we generate 10 meal plans and compute the average scores. It follows the methodology presented in "Diet Quality and Caloric Accuracy in AI-Generated Diet Plans: A Comparative Study Across Chatbots."[6]

All evaluations were conducted on held-out data not used in training.

Results

Main Results and Analysis

To evaluate the effectiveness of our proposed diabetes-specialized LLM in diabetes-specific multiple diabetes-related tasks, we conducted comprehensive experiments across five evaluation axes: QA, NLI, IE, summarization, and generation. Our model is compared with prior biomedical LLMs, including MedAlpaca 7B, PMC-LLaMA, ChatDoctor, AlpacaCare LLaMA2 7B, BioMistral, and Me-LLaMA. As shown in Table 1 and Table 2, our method consistently outperformed baseline models in most metrics, achieving the highest average performance across tasks.

In QA tasks (MedQA-USMLE, MedMCQA, PubMedQA), our model achieved scores of 0.44, 0.53, and 0.78 respectively, outperforming all baselines. The improvement is especially notable in PubMedQA (+0.23 over the next best model), highlighting the model's enhanced capability in diabetes-specific factual retrieval. For NLI (MedNLI), while MedAlpaca 7B slightly outperforms our model (0.55 vs. 0.39), its advantage appears to stem from exposure to a larger number of entailment/contradiction instructions. In contrast, our dataset includes only about 30 such examples, many of which were replaced during IFD-based filtering due to insufficient learning value. Nonetheless, our model maintains competitive NLI performance through a broader set of reformulated instructions emphasizing critical reasoning.

In IE (ADE Corpus), our model yields 0.39 accuracy, tying with the best baseline, BioMistral. More notably, in generative tasks—especially medical dialogue summarization (NoteChat) and clinical answer generation (iCliniq)—our model achieved the top average scores (0.74 and 0.722, respectively) based on automatic evaluation. GPT-4-based coherence, completeness, and naturalness ratings show our model not only maintains fluency, but also delivers contextually accurate and user-appropriate responses.

Unlike general biomedical LLMs fine-tuned on broad medical corpora, our model is trained on a diabetes-specialized instruction dataset curated through reflection-based replacement. We apply Instruction-Following Difficulty (IFD) and reversed IFD (r-IFD) scores to refine instruction samples that are either trivial or semantically ambiguous, ensuring that retained instructions are informative and aligned with the learning capacity of the student model. Low-quality samples are refined using GPT-3.5 to improve clarity and instruction-following signal. This targeted data refinement enables our model to outperform biomedical baselines in both short-form factual queries (e.g., "What is insulin?") and long-form reasoning tasks (e.g., "Compare the postprandial glycemic effect of rice and quinoa"), demonstrating stronger alignment with diabetes-specific reasoning demands.

Furthermore, we observe that models such as AlpacaCare perform well in generation tasks due to their summarization-heavy training data. However, our performance is comparable or better despite using fewer but more relevant instructions. Our model slightly underperforms in NLI, as previously mentioned, due to the lack of entailment-focused prompts in its final training set. Nevertheless, the deliberate inclusion of more complex reasoning-based tasks in our instruction set allows the model to compensate through generalizable inference patterns.

These results collectively demonstrate that a targeted, reflection-based replacement strategy tailored to diabetes management significantly enhances model performance. The resulting model shows strong capability in understanding and handling diverse tasks within diabetes management, positioning it as a promising candidate for domain-specific LLM deployment in real-world healthcare settings.

Ablation Study

To investigate the effectiveness of each component in our instruction tuning pipeline, we conducted a stepwise ablation study starting from the LLaMA3.1 8B base model. As shown in Table 3, we sequentially introduced Seed data, IFD-based replacement, r-IFD-based replacement, and curriculum instruction tuning.

Our base model, fine-tuned without any instruction optimization, shows limited reasoning ability and tends to memorize factual content rather than follow instructions. By introducing Seed instructions, performance improves significantly, especially in QA and Generation tasks, suggesting that even weakly-structured supervision can steer the model's generalization.

Further improvement is observed when incorporating Instruction-Following Difficulty (IFD) filtering. This step replaces low-IFD samples with more challenging, instruction-focused prompts. The change is reflected in the model's enhanced completeness and naturalness in both summarization and answer generation tasks.

The next replacement step, based on Reversed IFD (r-IFD), replaces overly deterministic instruction–response pairs with

Table 1. Performance comparison of instruction-tuned biomedical LLMs on QA, NLI, and IE tasks using diabetes-related queries. Our LLaMA3.1-based model achieves the highest average performance across all tasks.

Dataset	MedQA-USMLE	MedMCQA	PubmedQA	MedNLI	ADE_corpus	Avg.	Release Date	Base Model	Sample Size
MedAlpaca 7B	0.34	0.28	0.05	**0.55**	0.31	0.306	2023-04-14	LLaMA2	160K / 2.7K
pmc-llama	0.20	0.14	0.40	0.13	0.02	0.178	2023-04-27	LLaMA2	4.8M + 202M / –
chatdoctor	0.18	0.17	0.37	0.21	0.21	0.228	2023-05-24	LLaMA2	100K / 4.5K
Alpacare LLaMA2 7B	0.09	0.35	0.60	0.13	0.22	0.278	2023-10-23	LLaMA2	52K / 5.3K
biomistral	0.36	0.42	0.78	0.48	0.35	0.478	2024-02-14	Mistral	3B tokens / –
me-llama	0.36	0.33	0.26	0.23	0.31	0.298	2024-06-05	LLaMA3	192B tokens + 214K / –
LLaMA3.2 3B (Zero-shot)	0.24	0.30	0.20	0.07	0.29	0.220	2024-09-25	LLaMA3.2	–
LLaMA3.2 3B (Instruction)	0.29	0.31	0.78	0.25	0.34	0.394	–	LLaMA3.2	15.2K / 15.2K
LLaMA3.1 8B (Zero-shot)	0.34	0.30	0.55	0.21	0.30	0.340	2024-07-23	LLaMA3.1	–
LLaMA3.1 8B (Ours)	**0.44**	**0.53**	**0.78**	0.39	**0.39**	**0.506**	–	LLaMA3.1	15.2K / 15.2K

Table 2. Evaluation results for Medical Dialogue Summarization (NoteChat) and Medical Answer Generation (ICliniq). Human metrics (GPT-4) include coherence, completeness, and naturalness. Bold values represent the highest performance in each column.

Dataset	Coherence	Completeness	Naturalness	BLEURT	BERT score	Avg.
Medical Dialogue Summarization (NoteChat)						
MedAlpaca 7B	0.772	0.47	0.81	**0.72**	0.61	0.676
Pmc-llama	0.013	0.0	0.13	0.55	0.50	0.239
chatdoctor	0.405	0.315	0.485	0.41	0.43	0.409
Alpacare llama2 7B	0.802	0.675	0.847	0.67	0.62	0.723
biomistral	0.693	0.682	0.82	0.52	**0.73**	0.689
Me-llama	0.755	0.645	0.835	0.55	0.64	0.685
LLaMA3.2 3B Zero shot	0.32	0.235	0.335	0.41	0.34	0.328
LLaMA3.2 3B instruction	0.817	0.667	0.863	0.61	0.54	0.699
LLaMA3.1 8B Zero shot	0.355	0.287	0.382	0.35	0.33	0.341
LLaMA3.1 8B Instruction (Ours)	**0.823**	**0.693**	**0.863**	0.67	0.65	**0.74**
Medical Answer Generation (ICliniq)						
MedAlpaca 7B	0.63	0.45	0.7	0.56	0.61	0.59
Pmc-llama	0.245	0.162	0.38	0.60	0.62	0.401
chatdoctor	0.62	0.453	0.698	0.57	0.62	0.592
Alpacare llama2 7B	0.827	**0.665**	**0.935**	0.56	0.44	0.685
biomistral	0.502	0.312	0.613	0.55	0.52	0.499
Me-llama	0.68	0.483	0.798	0.60	0.67	0.646
LLaMA3.2 3B Zero shot	0.745	0.573	0.9	0.56	0.47	0.65
LLaMA3.2 3B instruction	0.775	0.603	0.907	**0.69**	0.50	0.695
LLaMA3.1 8B Zero shot	0.588	0.44	0.645	0.56	0.55	0.557
LLaMA3.1 8B Instruction (Ours)	**0.83**	0.655	0.925	0.57	**0.63**	**0.722**

more consistent and semantically reversible ones. This leads to increased instruction-following behavior. However, minor regressions are noted in QA and NLI performance—likely due to slight loss of fact-oriented clarity.

Finally, curriculum instruction tuning helps recover these drops by reordering training batches from low-IFD (easy) to high-IFD (challenging) instructions. The curriculum phase stabilizes the learning process and leads to the highest average performance across all metrics, especially in instruction-following and response naturalness.

Our analysis shows that the model transitions from a simple pattern-matching student model to a high-quality instruction follower. This is evidenced by verb–noun usage trends Figure. 2, which reveal a shift from response-oriented verbs (e.g., answer, give) to reasoning-driven verbs (e.g., analyze, tailor, design). Importantly, because we selectively applied replacements only to samples with IFD ≤ 0.45 and r-IFD ≥ 0.98, the task diversity of the instruction dataset is preserved.

Although our model does not outperform Medalpaca in NLI, and Alpacare in generation (which are both domain-specialized in those tasks), it achieves competitive or superior results across most tasks and shows the most balanced performance. Notably, our model maintains strong QA accuracy while exhibiting a marked improvement in instruction comprehension and reasoning capabilities, validating the benefits of targeted instruction remodeling and curriculum integration.

Evaluation of diabetes-friendly dietary recommendations

Table 4. Evaluation of natural language generation using GPT-4 scores for coherence, completeness, and naturalness.

Model	Coherence	Completeness	Naturalness
GPT-4	0.847	0.930	0.972
Ours	0.825	0.915	0.965

Table 3. Incremental ablation analysis across QA, IE, NLI, Summarization, and Generation tasks. Metrics include human evaluation (GPT-4) and automated scores. Each step cumulatively integrates a new method: Seed data, IFD filtering, r-IFD refinement, and Curriculum Learning.

Dataset	QA			NLI	IE	Summarization (NoteChat)					Generation (ICliniq)					Improvement (%)
	MedQA	MedMCQA	PubmedQA	MedNLI	ADE	Coh	Comp	Nat	BLEURT	BERT	Coh	Comp	Nat	BLEURT	BERT	
LLaMA3.1 8B Zero shot	0.34	0.30	0.55	0.21	0.30	0.355	0.287	0.382	0.35	0.33	0.588	0.44	0.645	0.56	0.55	–
+ Seed	0.41	0.37	0.69	0.41	0.32	0.837	0.67	0.853	0.41	0.45	0.81	0.608	0.913	0.61	0.67	+45.97
+ IFD	0.47	0.52	0.72	0.40	0.34	0.78	0.71	0.875	0.52	0.48	0.835	0.65	0.927	0.56	0.63	+52.21
+ r-IFD	0.43	0.52	0.82	0.37	0.36	0.823	0.718	0.875	0.65	0.60	0.808	0.635	0.883	0.63	0.57	+56.65
+ Curriculum Instruction Tuning	0.44	0.53	0.78	0.39	0.39	0.823	0.693	0.863	0.67	0.65	0.83	0.655	0.925	0.57	0.63	+59.03

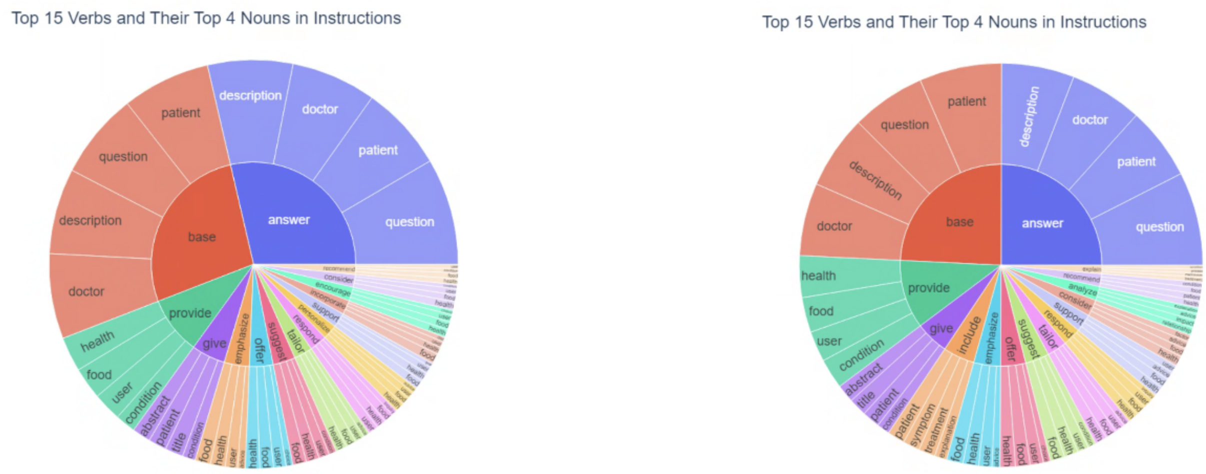

Figure 2. Instruction characteristics before and after reflection-based replacement using IFD and r-IFD. Post-replacement data emphasizes reasoning and task diversity, promoting deeper instruction-following ability beyond QA.

To further examine the real-world applicability of our model in the domain of diabetes management, we evaluate the quality of diet plans generated by our model in comparison to GPT-4. Specifically, both models were tasked with generating ten full-day meal plans for diabetic individuals. The generated outputs were evaluated from two perspectives: (1) natural language generation quality assessed by GPT-4-based metrics, including coherence, completeness, and naturalness, and (2) nutritional soundness based on the Diet Quality Index-International (DQI-I).

These results demonstrate that our model is capable of generating nutritionally appropriate and health-conscious meal plans suitable for diabetic management.

Overall, while GPT-4 demonstrated superior text generation quality, as shown in Table 4, our model produced meal plans better tailored to clinical needs, with higher DQI-I scores, as shown in Figure 3. These findings validate the efficacy of our domain-specific tuning approach in prioritizing nutritional integrity over general-purpose linguistic fluency.

Reflection-based Evaluation Criteria

To select high-quality instruction–response pairs, we applied empirical filtering thresholds using the Instruction Following Difficulty (IFD) and its reversed-IFD (r-IFD). IFD is defined as the ratio of conditional perplexity $p(y|x)$ to unconditional perplexity $p(y)$, while r-IFD is defined as $p(x|y)/p(x)$, highlighting how much the instruction guides the generation of the response and vice versa.

Fig. 4 illustrates the distribution of IFD and r-IFD scores for our instruction dataset. Based on this visualization, we set the filtering thresholds to:

These thresholds were chosen empirically by inspecting the data distribution in log-scale scatter plots. Samples in the bottom-left region of the IFD plot and top-right region of the r-IFD plot were considered informative and coherent. This dual-filtering mechanism allowed us to retain a diverse set of samples while reducing noisy or weakly-instructive pairs.

Limitations

While the proposed IFD and r-IFD metrics proved effective in filtering and substituting valuable instruction–response pairs, they are based on empirically determined thresholds. Specifically, we set filtering boundaries (e.g., IFD $\leq$ 0.45 and r-IFD $\geq$ 0.98) by visually inspecting the distribution of conditional perplexity across samples (Figure. 4). Although this empirical tuning provides practical utility, it lacks a theoretical justification or adaptive mechanism that could generalize across domains or datasets.

Moreover, the current filtering strategy may overlook valuable borderline samples that do not meet strict IFD/r-IFD cutoffs but could still contribute to diverse and instructional learning. Future work may explore adaptive thresholding techniques or probabilistic modeling to dynamically calibrate these metrics for more principled data selection.

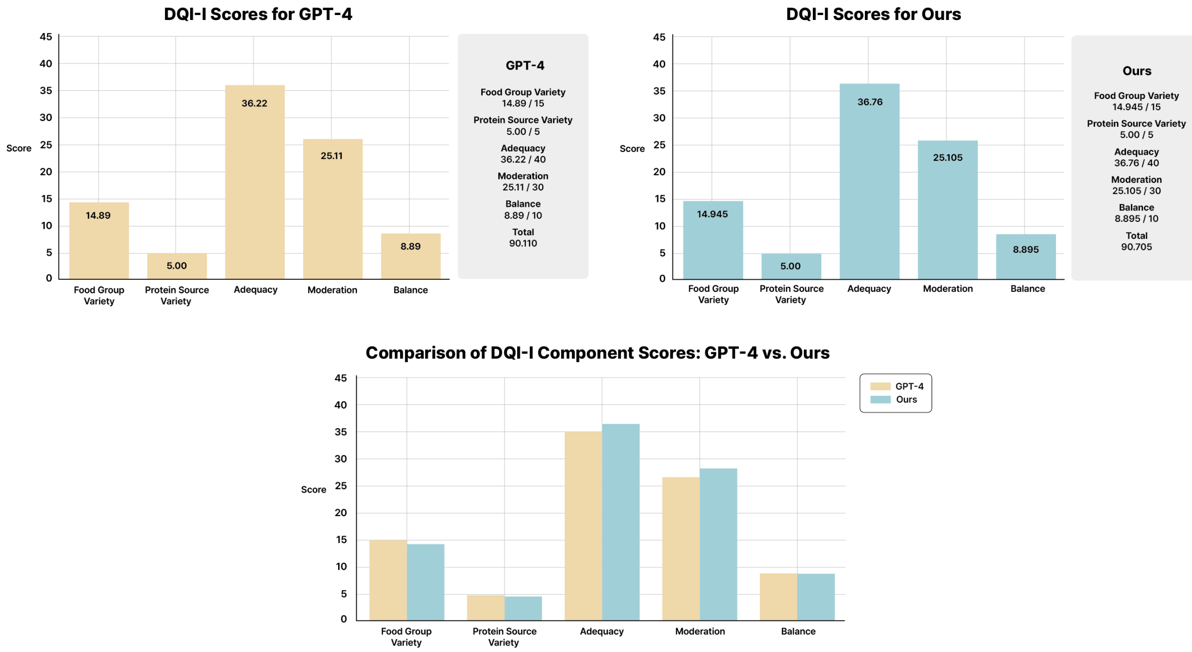

Figure 3. Nutritional quality assessment of meal plans based on DQI-I scores.

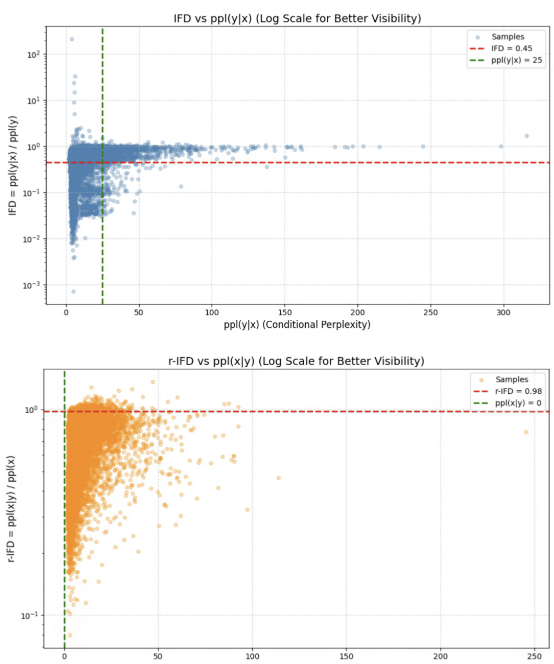

Figure 4. Samples selected for replacement lie below the IFD threshold (≤ 0.45) and above the r-IFD threshold (≥ 0.98).

Conclusion

This study presents a domain-specialized instruction-tuned large language model for diabetes management, addressing the limitations of general biomedical LLMs in handling fine-grained, task-specific instructions. While models such as BioBERT and BioGPT exhibit broad medical coverage, they fall short in delivering accurate and context-aware responses to diabetes-related queries, due in part to domain underrepresentation and catastrophic forgetting during adaptation.

To overcome this, we propose a reflection-based replacement pipeline leveraging Instruction-Following Difficulty (IFD) and reverse IFD (r-IFD) scores. These metrics enable the filtering and substitution of instruction–response pairs that are both pedagogically valuable and contextually feasible. We further incorporate a curriculum instruction tuning framework that introduces instructional complexity in stages, allowing the model to progressively acquire specialized diabetes knowledge while retaining previously learned capabilities.

Through comprehensive evaluations on QA, IE, NLI, summarization, and generation tasks, our model consistently outperforms existing biomedical LLMs in diabetes-focused evaluations. Notably, it demonstrates superior dietary planning capability as measured by DQI-I, and maintains competitive natural language generation quality against GPT-4.

Our findings highlight the potential of model-centered, domain-aware adaptation strategies for transforming general LLMs into expert systems for specific clinical contexts. Future work will explore the extension of this framework to other multimodal settings, such as integrating continuous glucose monitoring data and visual food recognition for real-time dietary guidance.

Key words: instruction tuning; large language models; question answering; natural language inference; information

extraction; text generation; multi-task learning.

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